STAFF REVIEW · H446 285 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Yes, routing decisions and changing network conditions can lead to different paths.

2. A reliable, ordered byte stream between endpoints.

3. Deny; AND requires both conditions true.

4. It permits needed service connections, but exposes that service to network-originated attacks or probing.

I Do — source worked steps

1. 443 — An inbound TCP connection destined for port 443 matches the allow rule and is permitted.

2. 22 — An inbound TCP connection destined for port 22 matches the deny rule and is blocked.

3. Scope — An allow rule permits traffic; it does not prove that an application or connection is harmless.

We Do — guided CHECK

1. Blocked resource
If the proxy's policy denies the destination, it does not forward the request to that site; the student receives a blocking response.

You Do — source mark schemes

1. [2] No, the connection is blocked [1], because inbound TCP destination port 22 matches the deny rule [1].

2. [4] Accept any two appropriately explained functions. Examples: apply browsing rules [1] to deny disallowed destinations before contacting them [1]; cache suitable responses [1] to reduce repeat external downloads [1]; or log request metadata [1] to support authorised monitoring or troubleshooting [1]. Maximum two functions × 2 marks.

3. [6] Allowed applications can contain exploitable vulnerabilities [1], with a relevant explanation such as an unpatched service being attacked on an allowed port [1]. Permitted traffic can also contain phishing or malicious content [1], which may still reach a browser or user [1]. Identify two additional safeguards with purpose, e.g. patching to close known flaws [1] and strong authentication or monitoring to reduce misuse or detect incidents [1]. Credit other valid distinct risks and defences.

Exit Ticket — source answer

1. False. A proxy acts as an intermediary for requests; a firewall enforces rules on traffic.